Pre-registered Monte Carlo protocols for the outer Soddy–Koide numerical observation

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Abstract

This document is the companion methods note to Brilliant (2026), *A Monte Carlo null-model test of an outer-Soddy completion of the Koide lepton triple*. It specifies in full the four Monte Carlo tests used to characterize the numerical observation $\mathcal{F}^2 \approx m_s^{\overline{\mathrm{MS}}}(\mu_\star)$, gives the underlying protocols, and reports the complete numerical output. The numerical observation itself predates the Monte Carlo design; the pre-registration described below applies only to the analysis protocols used to characterize that observation, not to the discovery of the observation itself. All analysis choices — prior shapes, sample sizes, hit criteria, seeds, cutoffs — were committed to the repository before any Monte Carlo was executed, with the commitment time-stamped in the version-pinned ledger.

The four tests address distinct null hypotheses: (A) random lepton-like spectra under four priors drawn from the flavor-model literature; (B) measurement-noise parametric bootstrap of the PDG 2024 charged-lepton inputs; (C) enumeration of 24 pre-specified lepton functions of comparable algebraic complexity; and (D) an exploratory temporal trajectory across the FLAG 2013–2024 $m_s^{\overline{\mathrm{MS}}}(2\ \mathrm{GeV})$ series. We report: under the primary log-uniform prior, a random-spectrum hit fraction of 0.340% [0.236, 0.475], stable within a factor of 2.4 across three prior shapes; of the 24 pre-specified lepton functions, two fall within 1σ of $m_s(\mu_\star)$ on the observed leptons, and those two are algebraically identical under exact Koide; and, as descriptive context, the residual is consistent with the 1σ band at every FLAG compilation between 2013 and 2024, with a coincidence-null Monte Carlo frequency of 0.26% for the all-five-epochs hit pattern under a specified prior on m_s^{true}. A technical observation, recorded for completeness, is that the canonical Yukawa-anarchy prior places only $\approx 2.5\%$ of its draws on the physical Koide sheet, so the random-spectrum hit fraction under that prior is conditioned on a small fraction of the prior mass.

None of the four tests is a hypothesis test, and the numerical values above are conditional frequencies under the specified constructions rather than p-values. They characterize the observation's behavior under stated null models and provide the reproducible numerical content for Section 5 of the main paper.

Companion paper: Brilliant (2026), "A Monte Carlo null-model test of an outer-Soddy completion of the Koide lepton triple" [1].
Repository: https://github.com/AndBrilliant/fermion_mass_mc
Keywords: Koide relation; Descartes circle theorem; Monte Carlo calibration; flavor physics; strange-quark mass; look-elsewhere effect.

1 Scope and motivation

The companion paper [1] reports a numerical observation: the outer Soddy curvature of the Descartes configuration whose three input curvatures are the charged-lepton mass square roots evaluates to

$$\mathcal{F}^2 \;=\; (e_1 - 2\sqrt{e_2})^2 \;=\; 95.1134\ \mathrm{MeV} \tag{1}$$

at PDG 2024 [8] central lepton masses, where e_k denote elementary symmetric polynomials in $\sqrt{m_\ell}$. The FLAG 2024 [9] $N_f = 2{+}1{+}1$ average for $m_s^{\overline{\mathrm{MS}}}$ at 2 GeV, 93.44 ± 0.68 MeV, evolved by

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four-loop QCD running [10] to the lepton-sum scale $\mu_\star = m_e + m_\mu + m_\tau = 1883.10$ MeV, gives

$$m_s^{\overline{\text{MS}}}(\mu_\star) \;=\; 95.075 \pm 0.692 \text{ MeV}. \tag{2}$$

The residual is

$$\mathcal{F}^2 - m_s^{\overline{\text{MS}}}(\mu_\star) \;=\; +0.038 \text{ MeV} \;\approx\; +0.055\,\sigma_{m_s}, \tag{3}$$

and is the subject of the tests below.

The main paper's Section 5 reports the Monte Carlo output at the level of tables and interpretations; the present document reports the underlying protocols and the full numerical output. Readers interested only in the conclusions may consult the main paper; readers interested in reproducing or auditing the analysis should use this document together with the repository.

The Monte Carlo tests described here address *the single numerical observation in equation (3)*. They do not speak to any of the following adjacent claims, each of which, if tested, would require its own Monte Carlo: Rivero's "Koide waterfall" (strange-charm-bottom, charm-bottom-top, lepton-to-top chaining) [6]; the golden quark-lepton mass relation [7]; any statement about the theoretical origin of the Koide relation itself; or any compound significance across multiple mass-formula coincidences. The v3 look-elsewhere companion [2] addresses the (c, b, t) heavy-Koide combination using the same Monte Carlo framework but with independent pre-registration.

2 Summary of results

For readers who want the bottom line before the protocols, Table 1 collects the numerical output. Each row points to the section in which the underlying protocol is defined.

Table 1: Summary of the Monte Carlo output. Each row reports a conditional frequency under the stated construction, not a p-value. Section references point to the protocol specification and full numerical output in this document.

Test	Construction	Numerical output	Section
A	Random lepton-like spectrum (primary, log-uniform)	0.340% [0.236, 0.475]	§5
A	Random lepton-like spectrum (log-normal, anarchy, linear)	0.25–0.59%; fold 2.4×	§5
A	Anarchic Yukawa fraction on Koide sheet (technical)	2.5%	§5.3
B	PDG measurement-noise bootstrap (baseline)	$P_{\leq \text{obs}} = 5.2\%$	§6
C	24 pre-specified lepton functions, within-1σ count	2 of 24	§7
D	FLAG 2013–2024 trajectory, coincidence-null MC (exploratory)	$P_{\text{all } 5} = 0.26\%$	§8

No combined significance is computed across the four tests; they probe partly overlapping constructions of the same observation and their independence is not established.

3 Engine validation

All four tests share a common computational engine, specified below.

Soddy pipeline. For a triple (m_1, m_2, m_3), the Descartes quadratic

$$(k_1 + k_2 + k_3 + k_4)^2 \;=\; 2(k_1^2 + k_2^2 + k_3^2 + k_4^2) \tag{4}$$

with $k_i = \sqrt{m_i}$ has smaller-curvature root $k_4^- = e_1 - 2\sqrt{e_2}$, where $e_1 = \sum_i k_i$ and $e_2 = \sum_{i<j} k_i k_j$. The outer Soddy curvature squared is $\mathcal{F}^2 = (k_4^-)^2$; the lepton-sum scale is $\mu_\star = m_1 + m_2 + m_3$. On PDG 2024 central lepton values, $\mathcal{F}^2 = 95.1134$ MeV and $\mu_\star = 1883.10$ MeV, reproducing the main paper to four significant figures.

Running. $\overline{\text{MS}}$ mass running is implemented via the `rundec` Python wrapper [10] at four loops, with flavor-threshold matching. Starting from FLAG 2024 [9] $m_s^{\overline{\text{MS}}}(2\text{ GeV}) = 93.44 \pm 0.68$ MeV, we match α_s at m_b ($n_f = 5 \to 4$), run to 2 GeV at $n_f = 4$, and run to $\mu_\star = 1.88310$ GeV at $n_f = 4$. Sanity check: $\alpha_s(m_\tau) = 0.3196$, consistent with PDG 2024 four-loop values. The engine reproduces the main paper's $m_s^{\overline{\text{MS}}}(\mu_\star) = 95.075 \pm 0.692$ MeV to four significant figures; the quoted ± 0.692 MeV is the FLAG lattice input propagated through running, and is the σ used for the hit criteria below. The α_s and four-loop-truncation contributions are reported separately in the main paper's error budget (Section 5.1) and are not folded into this σ.

Koide manifold constraint. A feature of direct sampling on the Koide manifold is the constraint that the "minus" root of the Koide quadratic yields an ordered spectrum $m_1 < m_2 < m_3$. Writing $u_i = \sqrt{m_i}$, the Koide condition $Q = 2/3$ in the middle position gives

$$u_2 \;=\; 2(u_1 + u_3) - \sqrt{3\,(u_1^2 + 4u_1u_3 + u_3^2)}. \tag{5}$$

Requiring $u_1 < u_2 < u_3$ reduces, after elementary algebra, to the constraint

$$\frac{u_3}{u_1} \;>\; 4 + \sqrt{18} \;\approx\; 8.24, \qquad \Leftrightarrow \qquad \frac{m_3}{m_1} \;>\; (4 + \sqrt{18})^2 \;\approx\; 67.9. \tag{6}$$

Triples with $m_3/m_1 \le 67.9$ do not lie on the physical sheet of the Koide manifold under the minus-branch convention used here. This is a pure algebraic property of the manifold, independent of any physical input. The observed leptons satisfy it comfortably: $m_\tau/m_e = 3477$. Section 5.3 records a consequence of (6) for anarchic Yukawa priors.

Direct sampler. Given a prior on (m_3, r) with $r = m_1/m_3$, the sampler sets $m_1 = r\,m_3$ and solves (5) for u_2 analytically. No rejection sampling on Koide is used; the direct sampler is exact. Triples with $m_3/m_1 \le 67.9$ produce $u_2 \le u_1$ or u_2 imaginary and are rejected by a mathematical validity check; this is the only acceptance step.

The fraction of prior draws that land on the physical Koide sheet depends only on the prior's support. For the primary Donoghue log-uniform prior (§5), the Koide-sheet acceptance rate is $\approx 58\%$, narrowing to $\approx 25\%$ once the $\mu_\star > 1$ GeV cutoff is applied (Table 2); for the HSW log-normal prior, $\approx 11\%$; for the Yukawa-anarchy SVD prior, $\approx 2.5\%$. These rates are a property of the prior's placement relative to the Koide manifold and are reported as part of the MC output.

4 Scope of pre-registration

The numerical observation in equation (3) predates the Monte Carlo design. The pre-registration described here applies only to the analysis protocols used to characterize that observation, not to the discovery of the observation itself.

With that scope fixed: all protocols enumerated in the following four sections — prior shapes, seeds, sample sizes, hit criteria, cutoffs, and test statistics — were committed to the repository as the file `PRE_REGISTRATION.md` before any Monte Carlo code was executed on a production data path. The commit hash appears in the repository ledger [15]; the file itself reproduces verbatim the planned choices, written in ordinary prose. A short "amendments" section at the bottom of that file records the sole addition made post-commit: Test D (temporal trajectory) was added on the same day, before any Monte Carlo ran, at which point the still-empty `stub.sh` and `output.log` files in the repository confirmed that no execution had yet occurred. The amendment is timestamped.

The purpose of the pre-registration is to draw a clear line between analysis decisions made before the Monte Carlo output was seen and decisions made after. It does not make the output a hypothesis test; the output remains a set of conditional frequencies under the specified constructions.

5 Test A: Random-spectrum Monte Carlo

Protocol 1 (Null A)**.** For each of four pre-registered prior shapes, draw $N = 10{,}000$ Koide-valid triples using the direct sampler of §3. For each triple, compute $\mathcal{F}^2$ and $\mu_\star$, run $m_s^{\overline{\mathrm{MS}}}(2\text{ GeV}) = 93.44 \pm 0.68$ MeV to the triple's $\mu_\star$, and count a "hit" if $|\mathcal{F}^2 - m_s^{\overline{\mathrm{MS}}}(\mu_\star)| < \sigma_{m_s}(\mu_\star)$. Apply the cut $\mu_\star > 1$ GeV (matches main paper §5.4). Pre-registered seeds 2026, 2027, 2028, 2029. Report the hit fraction with exact Clopper–Pearson 95% confidence interval.

5.1 Prior specifications

Each prior factorizes as a distribution on the heaviest mass m_3 and a conditional distribution on $r = m_1/m_3$, with m_2 then fixed by Koide via (5).

A1: Donoghue log-uniform (primary). m_3 is drawn log-uniformly on [1 MeV, 200 GeV]; r is drawn log-uniformly on $[10^{-3}, 10^{-1}]$. Physical motivation: Donoghue [11] observed that the charged-fermion spectrum is approximately log-uniform over six decades, with no apparent preference for any scale. This is the natural "uninformative" prior under that observation and serves as the primary result.

A2: Hall–Salem–Watari log-normal. m_3 is drawn log-normally centered on the geometric mean of the Standard-Model charged-fermion masses (median 0.1 GeV) with $\log\sigma = 2.5$; r is drawn log-normally with median 0.01 and $\log\sigma = 1.5$. Physical motivation: Hall, Salem, and Watari [12] argued for log-normal distributions of fermion masses arising from anthropic considerations on single-field inflation with a log-normal Yukawa distribution. This prior probes whether the random-spectrum output depends on heavy-tailed vs Gaussian-on-log placement of mass.

A3: Yukawa-anarchy SVD. m_3 is drawn log-uniformly as in A1; r is drawn as the ratio of minimum to maximum singular value of an i.i.d. $\mathcal{N}(0,1)$ 3×3 matrix. Physical motivation: Hall, Murayama, and Weiner [13] introduced anarchy as the hypothesis that Yukawa matrices are random structureless $O(1)$ matrices. For fermion masses, this gives singular-value ratios as mass ratios. de Gouvêa and Murayama [14] develop the statistical consequences.

A4: linear-uniform (stress test). m_3 is drawn linearly uniform on [1 MeV, 200 GeV]; r is drawn linearly uniform on $[10^{-3}, 10^{-1}]$. This prior places almost all weight on $m_3 > 10$ GeV, pushing $\mathcal{F}^2$ into the multi-GeV range far from m_s. A4 is not a physics proposal; it is included to bound the sensitivity of the hit fraction to a worst-reasonable-case prior choice.

5.2 Results

Table 2 reports the hit fractions and exact Clopper–Pearson 95% intervals.

The primary result is $p_{\mathrm{A1}} = 0.340\%$ [0.236, 0.475] at 95% confidence. Across the three physics-motivated priors (A1 log-uniform, A2 log-normal, A3 anarchy), the hit fraction varies by a factor of 2.4. Including the linear-uniform stress test widens this to a factor of 29.5, driven by A4's concentration of m_3 above 10 GeV, for which the typical residual is 5168 MeV — six orders of magnitude from the hit band and dominated by the prior's preference for mass scales where the observation cannot hold.

Table 2: Test A (Null A) results. The "Attempts" column gives the number of prior draws required to reach $N = 10{,}000$ Koide-valid triples; the "Acceptance rate" column is the corresponding fraction (Koide-valid triples with $\mu_\star > 1$ GeV, divided by attempts). Seeds pre-registered as 2026–2029 for A1–A4 respectively.

Prior	Attempts	Samples	Acceptance rate	Hit fraction	95% CI
A1 Donoghue log-uniform (primary)	39,364	10,000	25.4%	0.340%	[0.236, 0.475]
A2 Hall–Salem–Watari log-normal	90,464	10,000	11.1%	0.590%	[0.449, 0.760]
A3 Yukawa-anarchy SVD	406,654	10,000	2.46%	0.250%	[0.162, 0.369]
A4 linear-uniform (stress)	72,493	10,000	13.8%	0.020%	[0.002, 0.072]

The residual distribution is highly non-Gaussian: for A1 the median absolute residual is 665 MeV, and for A4 it is 5168 MeV. The hit fractions are tail events, not central ones. Random-spectrum pseudo-universes generically produce $\mathcal{F}^2$ values far from $m_s(\mu_\star)$; the small hit fractions quantify the rarity with which they land close.

5.3 Technical remark on the anarchy prior

Prior A3's low Koide-compatibility rate (2.5%) is a property of the Koide manifold under anarchic priors, independent of the outer-Soddy observation. The singular-value ratio distribution of i.i.d. Gaussian matrices has median ≈ 0.14, well above the Koide threshold $r < 1/67.9 \approx 0.0147$ derived in (6). Consequently 97.5% of anarchic-Yukawa draws produce triples that do not lie on the physical Koide sheet under the minus-branch convention used here; they are rejected as mathematically inadmissible, not as statistically unlikely.

We record this for completeness and note that it conditions the interpretation of the A3 hit fraction: the A3 number in Table 2 is 0.25% among the 2.5% of draws that reach the manifold, so the unconditional rate is $\approx 0.006\%$. We do not assign independent evidentiary weight to this in the comparison of (3).

5.4 Sensitivity

Within the three physics-motivated priors the hit fraction varies by a factor of $2.4\times$. This is small compared to the ratio of the observation's residual (0.038 MeV) to the typical residual under any prior (200–5000 MeV). The numerical output — that the random-spectrum null places the observation at around 0.3–0.6% — is stable across the three physics-motivated shapes.

6 Test B: Measurement-noise parametric bootstrap

Protocol 2 (Null B)**.** Draw $N = 2{,}000$ noise realizations per variant. For each realization, perturb the PDG 2024 charged-lepton masses by the specified noise model, recompute $\mathcal{F}^2$ and $\mu_\star$, re-run m_s to the perturbed $\mu_\star$, and record the signed residual. Pre-registered seeds 2126, 2127, 2128, 2129. Report the fraction of draws with $|\text{residual}| \leq 0.038$ MeV ("$P_{\leq \text{obs}}$") and the fraction with $|\text{residual}| < \sigma_{m_s}$ (the paper-style hit rate).

6.1 Variants

Each variant specifies a noise model on (m_e, m_μ, m_τ):

- **B1 (baseline):** uncorrelated Gaussian noise, standard deviation = PDG 2024 quoted uncertainty per lepton.
- **B2 (correlated):** multivariate Gaussian with off-diagonal correlation $\rho = 0.5$ across all three leptons. A proxy for a common-mode systematic.

- **B3 (tight):** uncorrelated Gaussian, standard deviation = 0.5× PDG 2024.
- **B4 (loose):** uncorrelated Gaussian, standard deviation = 2.0× PDG 2024.

6.2 Results

Table 3: Test B (Null B) results. $P_{\leq\mathrm{obs}}$ is the fraction of bootstrap draws with absolute residual ≤ 0.038 MeV (observed). Hit rate is the fraction with absolute residual $< \sigma_{m_s}$ (paper-style hit criterion). Signed mean and width are descriptive statistics.

Variant	σ scale	ρ	$P_{\leq\mathrm{obs}}$	Hit rate	Residual signed mean ± s.d. (MeV)
B1 baseline	1.0	0.0	0.0520	65.70%	$+0.029 \pm 0.712$
B2 correlated $\rho = 0.5$	1.0	0.5	0.0340	67.40%	$+0.008 \pm 0.698$
B3 tight	0.5	0.0	0.0910	94.80%	$+0.033 \pm 0.355$
B4 loose	2.0	0.0	0.0250	39.60%	$+0.004 \pm 1.338$

Two observations from Table 3:

1. The observed residual +0.038 MeV sits at the 5.2% percentile of the baseline B1 distribution, tightening slightly to 3.40% under correlated noise (B2). Both are small but neither is extreme. The observation is more precise than typical noise realizations produce, but not by a margin that would warrant a significance claim on its own.
2. The paper-style hit rate under B1 is 65.7%, consistent with the Gaussian expectation $\mathrm{erf}(1/\sqrt{2}) \approx 68.3\%$. This is a calibration check: the bootstrap is picking up the right amount of 1σ-band coverage, indicating that the engine correctly propagates the lepton uncertainties through $\mathcal{F}^2$ and through the $\overline{\mathrm{MS}}$ running.

The signed mean residual across all variants is consistent with zero, as it should be for a properly calibrated bootstrap around a small observed residual.

The purpose of Test B is primarily consistency: a bootstrap that placed the observed residual at the 0.1% tail would indicate a bug or an under-propagated uncertainty; one that placed it in the bulk would indicate that the observation is noise-compatible to begin with. Neither is the case. Test B does not on its own argue against a noise-level explanation.

7 Test C: Pre-specified comparison family of lepton functions

Protocol 3 (Comparison family)**.** Enumerate a pre-registered catalog of 24 functions of the three lepton masses, each of algebraic complexity comparable to $\mathcal{F}^2$. Evaluate each on the observed charged-lepton PDG 2024 central values and count hits within $1\sigma_{m_s}$ of $m_s^{\overline{\mathrm{MS}}}(\mu_\star)$. For each variant also run a smaller $N = 2{,}000$ Null-A pseudo-universe study using the primary A1 prior, to characterize the variant's per-function random-spectrum hit rate. Pre-registered seed 2226. Catalog frozen at commit time in `engine/algorithm_variants.py`; any post-run additions flagged as such. No post-run additions were made.

7.1 Rationale

The main paper's Section 3 argues on algebraic grounds that, once the Koide relation is read as a Descartes condition, the outer Soddy curvature $\mathcal{F}$ is the natural quantity for comparison with hadronic observables at the lepton-sum scale. The paper does not claim that no other function of the leptons hits $m_s(\mu_\star)$; it claims that the *choice* of the outer Soddy curvature is geometrically determined rather than retrospectively fitted. Test C quantifies this by enumerating similar-complexity alternatives.

The 24 variants are divided into several algebraic families: elementary symmetric polynomials in $\sqrt{m_\ell}$ and their squares and ratios; the Soddy curvatures $k_4^\pm$ squared; the "plus" root of the Koide quadratic squared; arithmetic, geometric, and harmonic means of the masses; pairwise $\sqrt{m_i m_j}$ products; simple differences $m_i - m_j$ and sums $m_i + m_j$; and two reduced-mass-like combinations. All 24 variants have algebraic complexity within a factor of two of $\mathcal{F}^2$'s, measured by the number of symbolic operations. They represent a defensible neighborhood in function space. The full list is in `engine/algorithm_variants.py`, frozen at the pre-registration commit.

7.2 Observed-lepton values

Table 4 gives the complete catalog of 24 variants and their values on the observed PDG 2024 lepton triple. This table mirrors Table 2 of the main paper; we include it here for self-containment and to expose the code-level names, which the paper omits for concision.

Table 4: All 24 comparison functions evaluated on PDG 2024 observed charged-lepton masses. "Code name" is the identifier in `engine/algorithm_variants.py` as frozen at the pre-registration commit. n_σ is the signed deviation from $m_s^{\overline{\mathrm{MS}}}(\mu_\star) = 95.075 \pm 0.692$ MeV. The two boldface rows are the only variants within 1σ, and are algebraically identical under exact Koide; rows 10, 11, 12 collide at observed-lepton precision (e_2 and $e_1^2/6$ coincide under exact Koide); rows 17 and 18 are literal duplicates; row 24 is dimensionally anomalous (mass dimension 3/2, not 1) and its value reflects the code's GeV-unit evaluation. All three features are retained per pre-registration.

#	Code name	Expression	Value (MeV)	n_σ
1	`outer_soddy_sq`	$\boldsymbol{(e_1 - 2\sqrt{e_2})^2 = \mathcal{F}^2}$	**95.113**	**+0.06**
2	`e1_minus_sqrt_p2`	$\boldsymbol{(e_1 - \sqrt{p_2})^2}$	**95.118**	**+0.06**
3	`inner_soddy_sq`	$(e_1 + 2\sqrt{e_2})^2$	9320.44	+13333.4
4	`koide_plus_root_sq`	$\left[2(\sqrt{m_1}+\sqrt{m_3}) + \sqrt{3}\sqrt{m_1+4\sqrt{m_1 m_3}+m_3}\right]^2$	25983.85	+37417.0
5	`e1_squared`	e_1^2	2824.66	+3945.1
6	`e1_squared_over_3`	$e_1^2/3$	941.55	+1223.4
7	`e1_sq_minus_p2`	$e_1^2 - p_2$ $(= 2e_2)$	941.56	+1223.4
8	`two_e2_minus_p2`	$2e_2 - p_2$	−941.54	−1498.2
9	`p2_minus_e2`	$p_2 - e_2$	1412.32	+1903.8
10	`e2_raw`	$e_2 = \sum_{i<j} \sqrt{m_i m_j}$	470.78	+543.0
11	`sqrt_e2_sq`	e_2 (duplicate enumeration)	470.78	+543.0
12	`sqrt_sum_sqrt`	$e_1^2/6$ $(= e_2$ under exact Koide)	470.78	+543.0
13	`sum_all`	$p_2 = \sum_i m_i$ $(= \mu_\star)$	1883.10	+2584.2
14	`arithmetic_mean`	$p_2/3$	627.70	+769.8
15	`geometric_mean`	$(m_1 m_2 m_3)^{1/3}$	45.78	−71.2
16	`harmonic_mean`	$3/\sum_i m_i^{-1}$	1.53	−135.2
17	`sqrt_m1_m3_prod`	$\sqrt{m_1 m_3}$	30.13	−93.9
18	`m_heavy_sqrt_times_light`	$\sqrt{m_3} \cdot \sqrt{m_1}$ (= row 17)	30.13	−93.9
19	`sqrt_m2_m3_prod`	$\sqrt{m_2 m_3}$	433.30	+488.8
20	`m1_plus_m2`	$m_1 + m_2$	106.17	+16.0
21	`m3_minus_m2`	$m_3 - m_2$	1671.27	+2278.1
22	`m3_over_m2_times_m2`	$m_2 m_3/(m_2 + m_3)$ reduced mass of (m_μ, m_τ)	99.73	+6.73
23	`m_mid_squared`	$m_2^2/(m_1 + m_3)$	6.28	−128.3
24	`fourth_root_prod_sq`	$\left[(m_1 m_2 m_3)^{1/4}\right]^2 = (m_1 m_2 m_3)^{1/2}$ (dim. 3/2)	9.80	−123.3

The two variants within 1σ are `outer_soddy_sq` and `e1_minus_sqrt_p2`, which are Proposition 1's two algebraically identical forms under exact Koide and differ at observed leptons by 0.005 MeV squared (the present Koide residual). The next-closest miss is row 22, the reduced mass of (m_μ, m_τ), at $+6.73\sigma$; the remaining 21 variants miss by $\geq 16\sigma$, with most clustered around either m_i (order 1 GeV, $+10^3\sigma$) or $m_i m_j$ (order 10^2–10^3 MeV^2, $+10^2$–$10^3\sigma$).

7.3 Pseudo-universe hit rates

For each variant, a smaller Null-A study ($N = 2{,}000$, prior A1, seed 2226) measures the variant's random-spectrum hit rate. The rationale is that some lepton-function families (e.g. the arithmetic mean) might have structurally small typical residuals and thus be favored in the Null-A test even if they do not hit on observed leptons. Table 5 shows that the per-function Null-A rates span a narrow range (0 to 0.45%), and the outer Soddy curvature's rate (0.20%) is comparable to the rates of variants that miss by 10^2–$10^3\sigma$ on observed leptons. The observed-lepton selectivity of Test C is therefore a property of *the leptons*, not of the function family: the ones that land happen to be the ones algebraically motivated by the Descartes reading.

Table 5: Per-function random-spectrum hit rates under Null-A prior A1, $N = 2{,}000$ Koide-valid triples per variant, seed 2226. Rates reported with exact Clopper–Pearson 95% intervals. Rows sorted by hit rate (descending); row numbers match Table 4. Only variants #1 and #2 hit the observed leptons at $\pm 1\sigma$; the distribution of Null-A hit rates across variants does not explain this selection.

#	Code name	Hits / N	Null-A hit rate	95% CI
16	`harmonic_mean`	9 / 2000	0.450%	[0.206, 0.853]
20	`m1_plus_m2`	8 / 2000	0.400%	[0.173, 0.787]
17	`sqrt_m1_m3_prod`	7 / 2000	0.350%	[0.141, 0.720]
18	`m_heavy_sqrt_times_light`	7 / 2000	0.350%	[0.141, 0.720]
15	`geometric_mean`	6 / 2000	0.300%	[0.110, 0.652]
23	`m_mid_squared`	6 / 2000	0.300%	[0.110, 0.652]
22	`m3_over_m2_times_m2`	5 / 2000	0.250%	[0.081, 0.582]
1	`outer_soddy_sq` *(hits observed)*	4 / 2000	0.200%	[0.055, 0.511]
2	`e1_minus_sqrt_p2` *(hits observed)*	4 / 2000	0.200%	[0.055, 0.511]
24	`fourth_root_prod_sq`	3 / 2000	0.150%	[0.031, 0.438]
Remaining 14 variants, all with 0/2000 *hits:*			0.000%	[0.000, 0.184]
	#3 `inner_soddy_sq`, #4 `koide_plus_root_sq`, #5 `e1_squared`, #6 `e1_squared_over_3`, #7 `e1_sq_minus_p2`, #8 `two_e2_minus_p2`, #9 `p2_minus_e2`, #10 `e2_raw`, #11 `sqrt_e2_sq`, #12 `sqrt_sum_sqrt`, #13 `sum_all`, #14 `arithmetic_mean`, #19 `sqrt_m2_m3_prod`, #21 `m3_minus_m2`.			

The outer Soddy curvature's Null-A hit rate (0.20%) is on the low side of the ten non-zero-rate variants and essentially identical to that of its Proposition 1 surrogate, as expected for two algebraically equivalent forms. The null-A rates do not discriminate the two hits from several variants that miss on observed leptons by $+10\sigma$ or more, confirming that the observed-lepton selectivity is not structurally baked into the enumerated function family.

8 Test D: Temporal trajectory

Protocol 4 (Temporal trajectory)**.** Load FLAG $m_s^{\overline{\mathrm{MS}}}(2\ \mathrm{GeV})$ central values and uncertainties from five successive FLAG reviews (2013, 2016, 2019, 2021, 2024). Evolve each to $\mu_\star$ using the same four-loop RunDec pipeline. For each epoch, compute the signed residual and residual/σ. Report: (i) the trajectory table; (ii) χ^2 against the null $m_s^{\mathrm{true}} = \mathcal{F}^2$ for the full five epochs and for a conservative three-epoch count treating 2019/2021/2024 as one correlated measurement; (iii) the inverse-variance weighted mean residual across epochs; (iv) a Monte Carlo coincidence-null construction with $N_{\mathrm{MC}} = 50{,}000$ draws of $m_s^{\mathrm{true}} \sim U(60, 130)$ MeV, reporting the fraction of MC draws in which $\mathcal{F}^2$ would land within 1σ at all five epochs. Pre-registered seed 2326. Test D is exploratory, not a hypothesis test; all outputs are descriptive conditional frequencies.

8.1 Rationale

A dimension of the data unused by Tests A, B, and C is time. As lattice precision on m_s has improved, the residual can be tabulated at each epoch, and its behavior recorded. We record this trajectory here as descriptive context for the observation; we do not use it to construct a significance test.

The charged-lepton masses are essentially stationary on the FLAG 2013–2024 timescale ($\sigma_{\mathcal{F}^2}^{\text{lepton}} \approx 0.01$ MeV is dominated by σ_{m_τ}, which has not meaningfully changed since the 2010s). $\mathcal{F}^2$ is therefore held fixed at its PDG 2024 value; only the lattice side evolves.

8.2 Historical inputs

The five FLAG $m_s^{\overline{\text{MS}}}(2\text{ GeV})$ values used are in `results/historical_masses.json`. For context:

- FLAG 2013: 93.8 ± 2.4 MeV, $N_f = 2{+}1$.
- FLAG 2016: 93.9 ± 1.1 MeV, $N_f = 2{+}1{+}1$.
- FLAG 2019, 2021, 2024: 93.44 ± 0.68 MeV, $N_f = 2{+}1{+}1$. (These three FLAG editions share most of their underlying lattice inputs; for this reason the "independent" count treats them as one.)

Over the 2013–2024 span, σ_{m_s} has narrowed by a factor of 3.5, and the central value has shifted by -0.36 MeV.

8.3 Trajectory

Table 6 reports the evolved-to-$\mu_\star$ trajectory.

Table 6: Test D trajectory. Each epoch's $m_s^{\overline{\text{MS}}}(\mu_\star)$ is the FLAG $m_s^{\overline{\text{MS}}}(2\text{ GeV})$ of that year, evolved by four-loop RunDec running. $\mathcal{F}^2 = 95.1134$ MeV held fixed at PDG 2024 charged-lepton values. Residual is $\mathcal{F}^2 - m_s^{\overline{\text{MS}}}(\mu_\star)$.

Epoch	N_f	$m_s(\mu_\star)$ (MeV)	σ_{m_s} (MeV)	Residual (MeV)	r/σ	Within 1σ?
FLAG 2013	2+1	95.441	2.442	−0.328	−0.134	yes
FLAG 2016	2+1+1	95.543	1.119	−0.430	−0.384	yes
FLAG 2019	2+1+1	95.075	0.692	+0.038	+0.055	yes
FLAG 2021	2+1+1	95.075	0.692	+0.038	+0.055	yes
FLAG 2024	2+1+1	95.075	0.692	+0.038	+0.055	yes

Three features of Table 6:

1. $\mathcal{F}^2$ lies within $1\sigma_{m_s}$ at every epoch, including the earliest two where the lattice error was $\sim 3\times$ larger.
2. r/σ stays near zero across the span: the earliest two epochs sit on the negative side of the band (-0.1 to -0.4σ), and the three most recent (correlated) epochs sit on the positive side ($+0.06\sigma$). The residual has migrated by a small fraction of σ, but has not drifted out of the band.
3. The residual in MeV is of order 0.1–0.4 MeV across the whole span, rather than growing or trending.

8.4 χ^2 against $m_s^{\text{true}} = \mathcal{F}^2$

Under the descriptive null $m_s^{\text{true}} = \mathcal{F}^2$, each epoch's residual is a draw from $\mathcal{N}(0, \sigma_i^2)$ in the limit where epochs are uncorrelated. The test statistic

$$\chi^2 \;=\; \sum_i \left(\frac{r_i}{\sigma_i}\right)^2 \tag{7}$$

would then follow $\chi^2_{N_{\text{indep}}}$.

Across all five epochs, $\chi^2 = 0.175$ on five degrees of freedom; p-value $= P(\chi^2_5 \geq 0.175) = 0.999$. The residuals are smaller than the null predicts — consistent with the null but sitting well inside the expected-fluctuation band.

With the conservative independent count (FLAG 2013, 2016, and FLAG 2019/2021/2024 collapsed to one), $\chi^2 = 0.169$ on three degrees of freedom, $p = 0.983$. Same qualitative conclusion.

Large p-values here point to compatibility with the descriptive null, not against it. The relevant statement is that the residuals are at most as large as random fluctuations under $m_s^{\text{true}} = \mathcal{F}^2$ would produce. No inferential claim is attached to this statistic; the independent-count assumption is unrealistic (see §8.7 below).

8.5 Weighted mean residual

The inverse-variance weighted mean residual across the five epochs is

$$\langle r \rangle_{\text{w}} \;=\; -0.022 \pm 0.372 \text{ MeV} \tag{8}$$

(-0.06σ from zero). This is consistent with zero, i.e. with $m_s^{\text{true}} = \mathcal{F}^2$. The same correlation caveat applies: the σ above treats the epochs as independent, which overstates the information content of the 2019/2021/2024 block.

8.6 Monte Carlo coincidence-null construction

As a descriptive complement to the χ^2 and weighted-mean statistics, we draw $m_s^{\text{true}} \sim U(60, 130)$ MeV (the range of historical determinations in the literature, including non-FLAG quenched results); for each epoch, sample the measured central value as $\mathcal{N}(m_s^{\text{true}}, \sigma_i^2)$; check whether $\mathcal{F}^2 = 95.113$ MeV lies within $\pm 1\sigma_i$ of the sampled central value; count the trajectory as a "hit" if it falls within the band at all five epochs. Run $N_{\text{MC}} = 50{,}000$ draws with pre-registered seed 2326.

The result is

$$P(\text{all 5 epochs hit} \mid m_s^{\text{true}} \sim U(60, 130)) \;=\; 0.26\%. \tag{9}$$

Per-epoch hit rates under the null are $7.1\%, 3.3\%, 2.0\%, 2.0\%, 1.9\%$ for FLAG 2013–2024 respectively; they decrease as σ narrows, as expected, because a random m_s^{true} off from $\mathcal{F}^2$ fails more visibly at the higher-precision epochs.

Under this specific coincidence-null construction, the observed multi-epoch hit pattern is uncommon. This figure is a conditional frequency under the stated construction; it is not a p-value and, in particular, does not account for correlations among FLAG 2019, 2021, and 2024, which share most of their underlying lattice inputs. We include it as descriptive context for the trajectory, not as an independent significance test.

8.7 Limits of Test D

Three limitations are stated explicitly.

Correlation. The three most recent FLAG compilations share most of their underlying lattice inputs. Treating them as independent overcounts the information they provide. The conservative three-epoch χ^2 and the MC construction retain the same qualitative behavior under this accounting, but the specific numerical values above should be read with this correlation in mind, not as the outputs of a fully independent five-epoch test.

FLAG is not m_s^{true}. The FLAG central values are not direct measurements of the true m_s but running averages with methodology-dependent systematics. Shifts in FLAG central values between editions can reflect improved systematics rather than pure measurement convergence. For Test D's purposes this is a benign uncertainty: $\mathcal{F}^2$ still has to land within whatever band the lattice community produces at each epoch, regardless of whether the band is centered on the true value.

Prior on m_s^{true}. The $U(60, 130)$ MeV range is wide but not arbitrarily so; it is the range of all historical determinations including pre-lattice, quenched, and early unquenched results. A narrower prior would produce a larger null frequency and would be defensible; a wider prior (say $U(0, 500)$ MeV) would produce a smaller null frequency. The width chosen is a pre-registered middle-ground choice, not one optimized for the result.

9 Combined interpretation

The four tests address different aspects of the observation:

- **A** — Random lepton-like spectrum: how rare the match is under a plausible physics prior on random triples. Result: 0.34% under the primary prior, stable within a factor of 2.4 across three physics-motivated priors, with the A3 rate conditioned on the small anarchic-Koide-compatibility fraction of §5.3.

- **B** — Measurement noise on the inputs: whether the observation is compatible with the PDG 2024 noise model. Result: observed residual at 5.2% percentile; most valuable as a calibration check on the engine.

- **C** — Algorithm selection: whether outer Soddy could have been cherry-picked from a larger catalog of lepton functions. Result: only 2 of 24 pre-specified variants hit the observed lepton combination; the two that do are algebraically identical under exact Koide; the observed selectivity is a property of the leptons themselves.

- **D** — Temporal trajectory: whether the match is stable across lattice-precision epochs. Result: $\mathcal{F}^2$ is consistent with the 1σ band at every FLAG compilation between 2013 and 2024; under the coincidence-null Monte Carlo construction of §8, the all-five-epochs hit pattern is uncommon (0.26%), with the caveats noted there.

The four tests probe different but partly overlapping constructions of the same observation. The numerical outputs above are conditional frequencies under the specified constructions, not p-values, and are not combined into a compound significance. No statement of independence across the tests has been established, and none is claimed.

The tests do not resolve the coincidence question. What they do is characterize the observation's behavior under a family of stated null models, in a form that supports the main paper's sensitivity analysis and is reproducible from the repository.

10 Reproducibility

Repository. All code, pre-registration, and numerical results are at https://github.com/AndBrilliant/fermion_mass_mc. The pre-registration commit is `5ab2b88` (2026-04-19); the production-run and audit-remediation commit is `3f09519` (2026-04-20). Both are accessible at the repository URL above.

Dependencies. Python 3.14.4; numpy 2.4.4; scipy 1.17.1; `rundec` (CRunDec Python wrapper, [10]). The `rundec` package must be installed separately from PyPI; see the repository README for instructions.

Seeds.

Test	Seeds
Null A, priors A1–A4	2026, 2027, 2028, 2029
Null B, variants B1–B4	2126, 2127, 2128, 2129
Test C	2226
Test D	2326

Reproduction. From the repository root:

```
python -m tests.validate_reproducibility    # smoke test, ~10 s
python -m tests.test_a_random_spectrum      # ~2--3 min, produces test_a_*.json
python -m tests.test_b_bootstrap            # ~30 s
python -m tests.test_c_algorithm_space      # ~30 s
python -m tests.test_d_temporal_convergence # ~5 s
```

On a mid-spec laptop, the core four-test sequence (A/B/C/D) completes in under five minutes; the full robustness suite including the filter-sensitivity and alternative-scales drivers completes in under ten minutes. All tables and numbers in this document are derivable from the JSON output files in `results/`, produced directly by the test scripts.

Pre-registration file. The complete pre-registration document `PRE_REGISTRATION.md` is included in the repository at the production-run commit. Anyone auditing the analysis should read that file first; the present document is written to be consistent with it.

Numerical output files.

- `results/test_a_random_spectrum.json`: full Null-A hit data for all four priors.
- `results/test_b_bootstrap.json`: full Null-B bootstrap data for all four variants.
- `results/test_c_algorithm_space.json`: 24-variant observed-lepton and pseudo-universe hit data.
- `results/test_d_temporal_convergence.json`: trajectory table, χ^2 outputs, weighted mean, and MC coincidence-null output.
- `results/historical_masses.json`: the FLAG 2013–2024 input values.

Engine version. Four-loop CRunDec via the `rundec` Python wrapper, with $\alpha_s(M_Z) = 0.1180$, $m_c(m_c) = 1.273$ GeV, $m_b(m_b) = 4.183$ GeV, $m_t = 173.0$ GeV. The engine validation (§3) shows agreement with the main paper's central values to four significant figures.

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